

MPact m6A Variant Scoring Pipeline

Standalone pipeline to score SNVs with the MPact model and estimate m6A effect size.

Quick Start (Copy-Paste)

```
cd /g/data/qq78/akanksha/m6A-snp/MPact_Scoring_Pipeline

# Create env (first time only)
python -m venv /g/data/qq78/akanksha/m6A-snp/.venv
source /g/data/qq78/akanksha/m6A-snp/.venv/bin/activate
pip install -r requirements.txt

# Run on bundled sample TSV
/g/data/qq78/akanksha/m6A-snp/.venv/bin/python score_mpact.py \
  --input mini_mixed.tsv \
  --output-tsv smoke_test_results.tsv \
  --fasta hg38.fa \
  --model-path model_window_501.h5 \
  --rediportal-gz TABLE1_hg38_v3.txt.gz \
  --scan-radius 5 \
  --batch-size 256
```

Bundled Paths (All In This Directory)

- Script: score_mpact.py
- Model: model_window_501.h5
- FASTA: hg38.fa
- FASTA index: hg38.fa.fai
- GTF: Homo_sapiens.GRCh38.110.gtf.gz
- A-to-I reference: TABLE1_hg38_v3.txt.gz
- Sample input TSV: mini_mixed.tsv
- Sample input VCF: mini_unannotated_test.vcf

Required annotation files: - Homo_sapiens.GRCh38.110.gtf.gz for strand inference - TABLE1_hg38_v3.txt.gz for A-to-I annotation

Inputs

Supported file types

- TSV

- VCF
- VCF.GZ

Required variant fields

TSV requires these columns (tab-separated):

#Chromosome	Position	Reference	Alteration
chr1	100000	A	G

VCF uses standard CHROM POS ID REF ALT columns.

Optional fields

- `transcript_id` / ENST to improve transcript-aware strand mapping.
- `strand` (if user-provided).
- Extra columns are preserved into output.

Core Run Commands

1. Run on your own TSV

```
cd /g/data/qq78/akanksha/m6A-snp/MPact_Scoring_Pipeline
/g/data/qq78/akanksha/m6A-snp/.venv/bin/python score_mpact.py \
--input /path/to/variants.tsv \
--output-tsv /path/to/predictions.tsv \
--fasta hg38.fa \
--model-path model_window_501.h5 \
--gtf Homo_sapiens.GRCh38.110.gtf.gz \
--rediportal-gz TABLE1_hg38_v3.txt.gz \
--output-plot /path/to/delta_histogram.png \
--scan-radius 5 \
--batch-size 512
```

2. Run on your own VCF

```
cd /g/data/qq78/akanksha/m6A-snp/MPact_Scoring_Pipeline
/g/data/qq78/akanksha/m6A-snp/.venv/bin/python score_mpact.py \
--input /path/to/variants.vcf.gz \
--output-tsv /path/to/predictions.tsv \
--fasta hg38.fa \
```

```
--model-path model_window_501.h5 \
--gtf Homo_sapiens.GRCh38.110.gtf.gz \
--rediportal-gz TABLE1_hg38_v3.txt.gz \
--scan-radius 5 \
--batch-size 512
```

3. Run on included sample (recommended first check)

```
cd /g/data/qq78/akanksha/m6A-snp/MPact_Scoring_Pipeline
/g/data/qq78/akanksha/m6A-snp/.venv/bin/python score_mpact.py \
--input mini_mixed.tsv \
--output-tsv sample_predictions.tsv \
--fasta hg38.fa \
--model-path model_window_501.h5 \
--gtf Homo_sapiens.GRCh38.110.gtf.gz \
--rediportal-gz TABLE1_hg38_v3.txt.gz \
--output-plot sample_delta_hist.png
```

Important Runtime Defaults

From current `score_mpact.py`: - **--gtf is required** for strand inference and uses `Homo_sapiens.GRCh38.110.gtf.gz` by default. - **--rediportal-gz is required** for A-to-I annotation and uses `TABLE1_hg38_v3.txt.gz` by default. - Default `--scan-radius`: 5 - Default `--batch-size`: 512 - Default `--input-chunk-size`: 10000 - VCF mode defaults to genic-only filtering (`--genic-only` is on).

Resume an interrupted run

If a run stops partway through, rerun the same command with `--resume`. You can also set `--resume-from-row` to continue from a specific input row and `--checkpoint-path` to point at the saved checkpoint JSON.

Example:

```
/g/data/qq78/akanksha/m6A-snp/.venv/bin/python score_mpact.py \
--input mini_mixed.tsv \
--output-tsv smoke_test_results.tsv \
--fasta hg38.fa \
--model-path model_window_501.h5 \
--gtf Homo_sapiens.GRCh38.110.gtf.gz \
--rediportal-gz TABLE1_hg38_v3.txt.gz \
--resume
```

If you want to keep intergenic VCF rows too, add:

--allow-nongenic

Output

Main output is a TSV with one row per scored candidate A-site.

Header dictionary (all output columns)

The output contains: - All normalized input variant columns (for example **#Chromosome**, **Position**, **Reference**, **Alteration**, and any additional input metadata columns) - Plus the MPact-generated columns below

Header	Meaning
strand_gencode	Final strand used for scoring (+ or -), inferred from GTF transcript/interval logic.
strand_gencode_source	Label of the GTF reference used (for example GRCh38.110).
score_type	Candidate class: disruption , creation , or context .
a_genomic_pos1	1-based genomic coordinate of the candidate A center that was scored.
snp_to_a_mRNA_offset	Signed offset from SNP to candidate A in oriented transcript coordinates.
ref_center_is_A	Whether reference 501-nt oriented sequence has A at center position.
alt_center_is_A	Whether alternate 501-nt oriented sequence has A at center position.
overlaps_AtoI_exact	True if candidate A exactly overlaps an indexed A-to-I site; otherwise False .
near_AtoI_5nt	True if nearest A-to-I site is within 5 nt; otherwise False .
near_AtoI_10nt	True if nearest A-to-I site is within 10 nt; otherwise False .
mpact_ref_score	MPact model score on reference-oriented 501-nt sequence.
mpact_alt_score	MPact model score on alternate-oriented 501-nt sequence.
mpact_ref_stoichiometry_pct	Reference score scaled to percent (mpact_ref_score * 100).
mpact_alt_stoichiometry_pct	Alternate score scaled to percent (mpact_alt_score * 100).
mpact_delta_stoichiometry_pct	Percent delta ((alt - ref) * 100).

Header	Meaning
<code>alt_center_A_destroyed</code>	True when ALT no longer has center A (not <code>alt_center_is_A</code>).
<code>mpact_delta_alt_minus_ref</code>	Raw effect size: <code>mpact_alt_score - mpact_ref_score</code> .
<code>mpact_abs_delta</code>	Absolute effect size: <code>abs(mpack_delta_alt_minus_ref)</code> .
<code>delta_zscore</code>	Z-score of raw delta against global delta distribution for the run.
<code>delta_p_two_sided</code>	Two-sided p-value computed from <code>delta_zscore</code> .
<code>ref10</code>	10-nt centered reference context slice from scored window.
<code>alt10</code>	10-nt centered alternate context slice from scored window.

Note: - `delta_zscore` and `delta_p_two_sided` are computed after chunk scoring using global delta mean/std over the temporary scored output.

Detailed interpretation guide

1) Input passthrough columns All input columns are preserved first in the output. This means original variant annotations (for example ClinVar labels, INFO-derived fields, cohort tags) stay attached to each scored candidate row.

Important behavior: - One input SNV can produce multiple output rows, because MPact scans candidate A centers around the SNV (`--scan-radius`). - The same input variant may therefore appear many times, each with a different `a_genomic_pos1` and possibly different `score_type`.

2) Strand and coordinate columns

- **strand_gencode:** The strand actually used for sequence orientation and scoring. This is the critical strand field to trust for downstream analysis.
- **strand_gencode_source:** The GTF build label used to infer strand. Useful for reproducibility and cross-run auditing.
- **a_genomic_pos1:** Candidate m6A-centered genomic coordinate (1-based).
- **snp_to_a_mRNA_offset:** SNP-to-center distance in transcript orientation.

Offset sign meaning: - Positive value: SNP is downstream of candidate A in the oriented transcript frame. - Negative value: SNP is upstream of candidate A in the oriented transcript frame. - Zero: SNP overlaps the candidate center position.

3) Event typing columns

- `ref_center_is_A` and `alt_center_is_A` describe whether the center base is A before and after applying ALT.
- `score_type` is derived from these booleans:
 - `disruption`: REF center is A and ALT center is not A
 - `creation`: REF center is not A and ALT center is A
 - `context`: REF center is A and ALT center is A
- `alt_center_A_destroyed` is simply `not alt_center_is_A` and is most informative for disruptive events.

4) **A-to-I proximity columns** These annotate known editing context around the candidate center: - `overlaps_AtoI_exact`: direct overlap - `near_AtoI_5nt`: local neighborhood overlap within 5 nt - `near_AtoI_10nt`: broader neighborhood overlap within 10 nt

Practical use: - Use `overlaps_AtoI_exact == True` for strict known-site overlap analyses. - Use `near_AtoI_5nt` or `near_AtoI_10nt` when testing local editing-environment enrichment.

5) **Score and delta columns** Core numeric fields: - `mpact_ref_score`: model score on REF sequence - `mpact_alt_score`: model score on ALT sequence - `mpact_delta_alt_minus_ref`: ALT minus REF (main direction-aware effect size) - `mpact_abs_delta`: absolute magnitude of effect

Percent fields are linear transforms of score fields: - `mpact_ref_stoichiometry_pct` = `mpact_ref_score * 100` - `mpact_alt_stoichiometry_pct` = `mpact_alt_score * 100` - `mpact_delta_stoichiometry_pct` = `mpact_delta_alt_minus_ref * 100`

Direction interpretation for `mpact_delta_alt_minus_ref`: - Negative: ALT decreases predicted m6A signal relative to REF - Positive: ALT increases predicted m6A signal relative to REF - Near zero: limited predicted effect for that candidate center

6) Statistical normalization columns

- `delta_zscore`: standardized delta across all candidate rows in the run
- `delta_p_two_sided`: two-sided p-value from that z-score

These are run-level normalized values, so they depend on the cohort/distribution in that specific run. If you compare different runs, compare carefully because z-score baselines can shift.

7) Local context columns

- `ref10` and `alt10` are short centered context strings from the full 501-nt windows.
- They are useful for quick motif-level sanity checks and visual inspection.
- They are not a replacement for full-window model inputs, but they are helpful for debugging and reporting.

8) Row-level interpretation example If a row has: - `score_type = disruption` - `mpact_delta_alt_minus_ref = -0.42` - `mpact_abs_delta = 0.42` - `delta_p_two_sided = 0.001`

Then that candidate center is predicted to show a strong ALT-driven loss of m6A signal, with relatively extreme effect size within the run distribution.

Optional histogram PNG is written when `--output-plot` is set.

Test command

```
cd /g/data/qq78/akanksha/m6A-snp/MPact_Scoring_Pipeline
/g/data/qq78/akanksha/m6A-snp/.venv/bin/python score_mpact.py \
  --input mini_mixed.tsv \
  --output-tsv smoke_test_results.tsv \
  --fasta hg38.fa \
  --model-path model_window_501.h5 \
  --scan-radius 5 \
  --batch-size 256
```

Observed result: - Exit code 0 - Output TSV created successfully - 5 lines total (header + 4 scored candidates)

HPC PBS Usage

Template script: `submit_mpact_scoring.pbs`

Before `qsub`, update: - `#PBS -P`, `queue`, `walltime`, `ncpus`, `mem`, `storage` - `INPUT_PATH`, `OUTPUT_DIR`, `FASTA`, `MODEL` - optional `REDIPORTAL_GZ`, `GTF`

Submit with:

```
qsub submit_mpact_scoring.pbs
```

Resume Note

The run is resumable with the updated PBS wrapper:

```
qsub /g/data/qq78/akanksha/m6A-snp/Scripts/run_gnomad_genic_snv_dtm6a501_scan20.pbs
```

The wrapper calls `score_mpact.py` with `--resume`, so already-scored chunk outputs in `external_validation/gnomad/chrom_chunks` are skipped and only missing chunks are processed.

The launcher now passes absolute paths for: - FASTA: `MPact_Scoring_Pipeline/hg38.fa`
- GTF: `MPact_Scoring_Pipeline/Homo_sapiens.GRCh38.110.gtf.gz` - REDI-portal: `MPact_Scoring_Pipeline/TABLE1_hg38_v3.txt.gz`

Troubleshooting

Missing FASTA index

```
samtools faidx /path/to/hg38.fa
```

samtools not found

```
module load samtools  
which samtools
```

Slow startup or no GPU on login node

TensorFlow may print CUDA warnings on CPU-only nodes. This is expected if no GPU is present, and CPU inference still runs.

Memory pressure

Use a smaller batch size:

```
--batch-size 256
```